

D827

Fundamentals of Information Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

At a 450-bed hospital, a faulty database script silently changed dosage values in 300 patient records. No data was disclosed and the system stayed online the whole time. The hospital recently earned a national accreditation. Which security goal was compromised?

- A. Integrity
- B. Non-repudiation
- C. Availability
- D. Confidentiality

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A server room is valued at \$200,000. A flood would destroy 25% of its value, and floods are expected once every five years. The building is 12 years old, and the IT manager holds two certifications. What is the annualized loss expectancy?

- A. \$40,000
- B. \$10,000
- C. \$250,000
- D. \$50,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A control costs \$8,000 a year. It lowers ALE from \$30,000 to \$12,000. What is the control's annual value to the organization?

- A. \$18,000
- B. \$4,000
- C. -\$10,000
- D. \$10,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A credit union with 14 branches installs cameras that record everyone entering its server room, and a supervisor reviews the footage weekly. What function does this control mainly serve?

- A. Detective
- B. Corrective
- C. Preventive, because cameras stop people from entering.
- D. Deterrent only; cameras can never detect anything.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which of the following is an administrative control?

- A. Disk encryption, since all security controls are technical
- B. A badge reader on the data center door
- C. An acceptable use policy that staff acknowledge each year
- D. A firewall rule blocking inbound traffic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

At a manufacturer with 900 employees, one accounts payable clerk can both add new vendors and approve payments to them in the company's cloud ERP system. What is the best fix?

- A. Add a stronger network firewall.
- B. Separate the duties so different people add vendors and approve payments.
- C. Remove the clerk's ERP access entirely.
- D. Move the ERP system back on premises.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A law firm with 60 attorneys requires a password and an answer to a security question at login and calls this multi-factor authentication. Which change would make it real MFA?

- A. Add a second PIN.
- B. Require longer passwords.
- C. Replace the security question with an authenticator app or hardware key.
- D. MFA is impossible without fingerprint readers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which pair is genuine two-factor authentication?

- A.** A smart badge and a PIN
- B.** A password and a PIN
- C.** A username and a password
- D.** A fingerprint and a face scan

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A school district with 30 schools relied only on a perimeter firewall. A laptop infected at a staff member's home was brought inside, and malware spread across the internal network. The school board just approved a new technology budget. Which improvement best reflects sound security design?

- A.** Spend the budget on new laptops for all 30 schools.
- B.** Add layers inside the perimeter: endpoint protection, network segmentation, patching, and monitoring.
- C.** A strong enough firewall makes internal controls unnecessary.
- D.** Buy a larger perimeter firewall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An online retailer that sells in three states and runs two warehouses buys a cyber insurance policy to cover the costs of a possible data breach. Which risk response is this?

- A.** Accept
- B.** Mitigate
- C.** Avoid
- D.** Transfer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A company cancels its planned mobile app because the security risks outweigh the expected benefit. Which risk response is this?

- A.** Transfer
- B.** Accept
- C.** Mitigate
- D.** Avoid

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

At a county office with 200 staff, an employee (1) types a username, (2) enters a password and a token code, (3) can open only the HR folders, and (4) has every file access logged. The office moved to a new directory service last year. Which step is authorization?

- A.** Step 1
- B.** Step 4
- C.** Step 3
- D.** Step 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

~~At a 450-bed hospital,~~ a faulty database script silently changed dosage values in 300 patient records. No data was disclosed and the system stayed online the whole time. ~~The hospital recently earned a national accreditation.~~ Which security goal was compromised?

Target word: which CIA element **Decides:** Values were changed; nothing exposed or offline.

A	Correct	Unauthorized change to data is an integrity failure.
B	Adjacent category	About proving who did an action, not whether data stayed accurate.
C	Confusion set	The system stayed online.
D	Confusion set	No one saw data they should not have.

2. Answer B Concept 2

A server room is valued at \$200,000. A flood would destroy 25% of its value, and floods are expected once every five years. ~~The building is 12 years old, and the IT manager holds two certifications.~~ What is the annualized loss expectancy?

Target word: ALE **Decides:** SLE = \$50,000; ARO = 0.2.

A	Wrong stem number	$\$200,000 \times 0.2$ uses the full asset value instead of the SLE.
B	Correct	$\$200,000 \times 0.25 = \$50,000$ SLE; $\times 0.2 = \$10,000$.
C	Inverted operation	$\$50,000 \div 0.2$ divides by the rate instead of multiplying.
D	Dropped element	That is the SLE; it drops the annual rate.

3. Answer D Concept 2 · transfer

A control costs \$8,000 a year. It lowers ALE from \$30,000 to \$12,000. What is the control's annual value to the organization?

Target word: solve for net control value **Decides:** ALE before – ALE after – control cost.

A	Dropped element	Leaves out the control's cost.
B	Wrong stem number	$\$12,000 - \$8,000$ uses the remaining ALE.
C	Inverted operation	Subtracts in the wrong direction; the control saves money.
D	Correct	$\$30,000 - \$12,000 - \$8,000 = \$10,000$.

4. Answer A Concept 3

A credit union ~~with 14 branches~~ installs cameras that record everyone entering its server room, and a supervisor reviews the footage weekly. What function does this control mainly serve?

Target word: control function **Decides:** Footage is reviewed after the fact.

A	Correct	Recording and review identify events after they happen.
B	Adjacent category	Corrective controls restore systems after an incident.
C	Confusion set	Cameras do not physically block entry; a lock would.
D	Absolute language	They may deter, but their review is detection.

5. Answer C Concept 3 • transfer

Which of the following is an administrative control?

Target word: reverse: from category to example **Decides:** Administrative = policies, procedures, training.

A	Absolute language	Controls also include administrative and physical types.
B	Confusion set	A physical control.
C	Correct	Policies and training are administrative.
D	Adjacent category	A technical control.

6. Answer B Concept 4

At a manufacturer ~~with 900 employees~~, one accounts payable clerk can both add new vendors and approve payments to them ~~in the company's cloud ERP system~~. What is the best fix?

Target word: principle and fix **Decides:** One person controls both steps of a sensitive process.

A	Adjacent category	A network control does not fix a conflict of duties inside the app.
B	Correct	Splitting the steps prevents one person from creating and paying a fake vendor.
C	Absolute language	Overcorrects; the clerk still needs access for one of the duties.
D	Camouflage	Hosting does not decide who can approve payments.

7. Answer C Concept 5

A law firm ~~with 60 attorneys~~ requires a password and an answer to a security question at login and calls this multi-factor authentication. Which change would make it real MFA?

Target word: what makes MFA **Decides:** Password and security question are both something you know.

A	Confusion set	Still something you know; same category.
B	Adjacent category	Strengthens one factor; does not add another.
C	Correct	Adds a something-you-have factor.
D	Absolute language	A something-you-have factor works too.

8. Answer A Concept 5 • transfer

Which pair is genuine two-factor authentication?

Target word: reverse: pick valid factor pair **Decides:** Factors must come from two categories.

A	Correct	Something you have plus something you know.
B	Confusion set	Both are something you know.
C	Dropped element	The username only identifies; there is one factor.
D	Confusion set	Both are something you are.

9. Answer B Concept 6

A school district ~~with 30 schools~~ relied only on a perimeter firewall. A laptop infected at a staff member's home was brought inside, and malware spread across the internal network. ~~The school board just approved a new technology budget.~~ Which improvement best reflects sound security design?

Target word: design principle to apply **Decides:** One layer failed and nothing inside stopped the spread.

A	Camouflage	New hardware without layered controls repeats the gap.
B	Correct	Defense in depth limits damage when one layer is bypassed.
C	Absolute language	This incident bypassed the firewall entirely.
D	Dropped element	Still one layer; the threat came from inside.

10. Answer D Concept 7

An online retailer ~~that sells in three states and runs two warehouses~~ buys a cyber insurance policy to cover the costs of a possible data breach. Which risk response is this?

Target word: name the risk response **Decides:** Insurance shifts financial impact to a third party.

A	Adjacent category	Acceptance means taking no action; buying insurance is an action.
B	Confusion set	Mitigation lowers likelihood or impact; insurance does neither.
C	Confusion set	The retailer still collects data; the activity continues.
D	Correct	Insurance moves the cost of the loss to the insurer.

11. Answer D Concept 7 · transfer

A company cancels its planned mobile app because the security risks outweigh the expected benefit. Which risk response is this?

Target word: reverse: from action to response **Decides:** The risky activity is not undertaken.

A	Confusion set	No third party takes on the cost.
B	Inverted operation	Acceptance means going ahead and living with the risk.
C	Adjacent category	No safeguards are added; the project is dropped.
D	Correct	Dropping the activity removes the risk.

12. Answer C Concept 8

At a county office ~~with 200 staff~~, an employee (1) types a username, (2) enters a password and a token code, (3) can open only the HR folders, and (4) has every file access logged. ~~The office moved to a new directory service last year.~~ Which step is authorization?

Target word: identify authorization **Decides:** Authorization sets what a proven identity may access.

A	Confusion set	Typing a username is identification.
B	Adjacent category	Logging activity is accounting.
C	Correct	Limiting access to HR folders is authorization.
D	Confusion set	Proving identity is authentication.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The CIA triad	Confidentiality = only authorized people see data; integrity = data is accurate and unaltered; availability = systems and data are usable when needed.	Calling every incident a confidentiality breach.	What went wrong: exposed, changed, or unreachable.
2	Quantitative risk (SLE, ARO, ALE)	$SLE = \text{asset value} \times \text{exposure factor}$; $ALE = SLE \times ARO$; a control is worth ALE before – ALE after – annual control cost.	Stopping at SLE or skipping the exposure factor.	An asset value, a percentage of loss, and a frequency per year.
3	Control categories and functions	Controls are administrative, technical, or physical, and they act as preventive, detective, or corrective.	Treating a camera as preventive or a policy as technical.	Whether the control stops, spots, or repairs — and whether it is people, tech, or physical.
4	Least privilege and separation of duties	Give each person only the access the job needs, and split critical steps so no one person can both start and approve a sensitive action.	Fixing a conflict of duties by removing all access or adding a network control.	One person can create and approve the same transaction.
5	Authentication factors	Multi-factor authentication needs factors from different categories: something you know, have, or are.	Counting two items from the same category as MFA.	Password plus security question, or password plus PIN.
6	Defense in depth	Layer independent controls so the failure of one layer does not expose the asset.	Relying on one strong perimeter control.	A single control failed and nothing else stopped the spread.
7	Risk responses	Avoid (stop the activity), mitigate (reduce likelihood or impact), transfer (shift cost, e.g., insurance), or accept (document and live with it).	Calling insurance mitigation or calling cancellation acceptance.	Insurance, contracts, canceled projects, or new safeguards.
8	Identification, authentication, authorization, accounting	Identification claims an identity, authentication proves it, authorization sets what it may do, accounting records what it did.	Using authentication and authorization interchangeably.	A sequence of login and access steps.

Course scope (public description): Fundamentals of Information Security lays the foundation for understanding terminology, principles, processes, and best practices of information security at local and global levels. It further provides an overview of basic security vulnerabilities and countermeasures for protecting information assets through planning and administrative controls within an organization.